



8 MW NONGKHAI WASTE TO ENERGY PROJECT IN THAILAND



Document Prepared by Kosher Climate India Private Limited

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1 PROJECT DETAILS

1.1 Summary Description of the Project

The project is 8 MW Nong khai nayu Waste to Energy Project by Nong khai nayu Co., Ltd. is located in Thailand. The Plant will utilize Refuse-Derived Fuel (“RDF”) as feedstock derived from the separation and sorting of Municipal Solid Waste (“MSW”) removing recyclable or potentially hazardous materials from the waste stream. The feedstock will be supplied from 100% RDF by weight primarily sourced from two (2) municipal waste plants which already have the Memorandum of Understanding (“MOU”).

The main purpose of this project activity is to generate clean form of electricity through waste to energy-based power generation. The project activity involves installation of a 8 MW installed capacity of WTE power project in Phon Sawang Sub-District, Mueang Nong Khai District, Nong Khai province, Thailand.

Project is owned and implemented by Nongkhainayu Co., Ltd. Project has started development and yet to be commissioned

<i>Project activity</i>	<i>Capacity (MW)</i>	<i>Commissioning Date</i>
<i>Nongkhainayu Co., Ltd</i>	8	Yet to be commissioned

Scenario existing prior to the implementation of project activity:

This is a green field project activity, there is no power plant operating in the project area before the construction of this project activity. In the absence of the project activity the equivalent amount of electricity would have otherwise been generated by the operation of grid connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the “Tool to calculate the emission factor for an electricity system”.

Baseline Scenario:

The methane emissions that are avoided due to the project activity have otherwise been emitted due to unmanaged landfills in the captive area where the MSW has been collected. As far as the electricity generated at the WTE power plant is concerned, as per the applicable methodology, a Greenfield power plant is defined as “a new renewable energy power plant that is constructed and operated at a site where no renewable energy power plant was operated prior to the implementation of the project activity”. As the project activity falls under the definition of a Greenfield power plant, the baseline scenario as per the applied methodology is the following:

If the project activity is the installation of a Greenfield power plant, the baseline scenario is electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the “Tool to calculate the emission factor for an electricity system”.

Hence, pre-project scenario and baseline scenario are the same.

Project activity on average reduces 44,165 tCO₂e amount of GHG emission reductions per year. Over the first crediting period of 7 years project activity reduces 309,154 tCO₂e emission reductions.

1.2 Sectoral Scope and Project Type

Sectoral Scope :01- Energy Industries (Renewable/non-renewable sources) and 13 –
Waste handling and disposal

Project Type : Municipal solid to energy project

1.3 Project Eligibility

This is a 8 MW small scale grid connected electricity generation project using *waste to energy technology* in the *Thailand* which is a *developing Asian country* as per the UN. Hence This project is eligible under the scope of the VCS program.

1.4 Project Design

The project has been designed to include a single installation of an activity.

Eligibility Criteria

The project is not a grouped project. Thus, this section is not applicable Project Proponent

1.5 Project Proponent

Organization name	Nongkhainayu Co., Ltd
Contact person	Sunee Mounghareon
Title	Manager
Address	223/61, Country Complex Tower A, 14 th Floor, Sanphawut Road, Bangna sub-district, Bangna district, Thailand 10260
Telephone	+91 28 7109 9209

Email	Sunee.m@supercorp.co.th
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1.6 Other Entities Involved in the Project

Organization name	Kosher Climate India Private Limited
Role in the project	Project Representative
Contact person	Vamsi Krishna
Title	Mr.
Address	Zee Plaza, No.1678, 27th Main Road, Sector 2, HSR Layout, Bengaluru, Karnataka 560102, India
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Email	vamsi@kosherclimate.com

1.7 Ownership

The project owner is Nongkhainayu Co., Ltd.. The business license of the project owner are the evidences for legislative right. Besides, the project allotment license, the equipment purchasing contract and the power purchase agreement are the evidences for the ownership of the plant equipment and power generating.

1.8 Project Start Date

30-December-2022 (Expected Commercial Operation Date)

1.9 Project Crediting Period

Project Owner has chosen the renewable crediting period of 7 Years and the first crediting period starts from 01/01/2023 to 31/12/2029

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

The estimated annual GHG emission reductions/removals of the project are:

- ☐ <20,000 tCO₂e/year
- ☒ 20,000 – 100,000 tCO₂e/year
- ☐ 100,001 – 1,000,000 tCO₂e/year
- ☐ >1,000,000 tCO₂e/year

In all other cases, indicate the scale of the project (project or large project) and the estimated annual GHG emission reductions or removals for the project crediting period.

Project Scale	
Project	√
Large project	

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
Year 2023	30771
Year 2024	37571
Year 2025	42565
Year 2026	46238
Year 2027	48913
Year 2028	50848
Year 2029	52247
Total estimated ERs	309,154
Total number of crediting years	7
Average annual ERs	44,165

1.11 Description of the Project Activity

The project is 8 MW Municipal Waste to Energy Project by Nongkhainayu Co., Ltd. is located in Thailand. The Plant will utilize Refuse-Derived Fuel (“RDF”) as feedstock derived from the separation and sorting of Municipal Solid Waste (“MSW”) removing recyclable or potentially hazardous materials from the waste stream. The feedstock will be supplied from 100% RDF by weight primarily sourced from two (2) municipal waste plants which are Tha Bo and Phon Pi Sai sorting waste source.

(1) steam generator (boiler), one (1) steam turbine generator with air cooled condensing system and all associated auxiliaries.

This heating value is matched with the information from financial model after converting to 14,235.80 kJ/kg at LHV while the expected fuel consumption assuming at 335 operation days

at the gross electrical power output of 8 MW would result in annual fuel consumption of 90,370 tons/year or 269.76 tons/day.

The projects are yet to be commissioned.

Project Location	Project Promoters	Installed Capacity (MW)	Commissioning Date
Nongkhainayu Project	Nongkhainayu Co., Ltd	8	Yet to be commissioned

The project will replace anthropogenic emissions of greenhouse gases (GHG's) estimated to be approximately 44,165 tCO_{2e} per year, thereon displacing estimated value of around 40 Million kWh/year amount of electricity from the generation-mix of power plants connected to the Thailand grid, which is mainly dominated by thermal/fossil fuel-based power plant. Over the first crediting period the project activity estimated to reduce 309,154 tCO_{2e} GHG emission reductions.

The main fuel source for this project is 380 tons of sorted solid waste that will be delivered to the project daily in accordance with the solid waste management agreement between Nong Khai Provincial Administrative Organization and Nongkhainayu Company Limited. MSW will be sent to separate the non-combustible part out to get the RDF product approximately 270 TPD that will be fed to the boiler as the feedstock of the Project.

Baseline Scenario

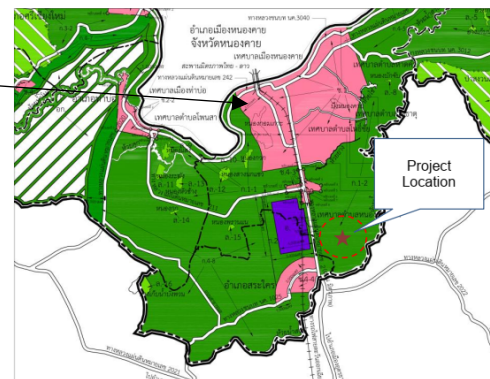
The scenario existing prior to the implementation of the project activity, is electricity delivered to the grid by the project activity that would have otherwise been generated by the operation of grid connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the **“Tool to calculate the emission factor for an electricity system”**. This is a green field project activity. There was no activity at the site of the project participant prior to the implementation of this project activity. Hence pre-project scenario and baseline scenario is the same.

As far as methane avoidance, the eastern part of Thailand, In 2016, available supply quantities of MSW in Thailand and specifically, in eastern Thailand were 5,444,713 tons and 1,644,261 tons, respectively. All the waste generated in these cities dumped in open landfills unscientifically, leading to acute shortage of land fill site near the city. The project activity involves the processing of municipal solid waste in to fluff and generation of power using the fluff as fuel in power boilers. The waste otherwise would have been dumped in unsanitary landfills thereby leading to methane emissions.

1.12 Project Location

The project is located in *the* Muang Nong Khai district, Nong Khai Province.

Capacity (MW)	Physical address	latitude	Longitude
8	No. 999 Moo 8, Phon Sawang sub-district, Muang Nong Khai district, Nong Khai Province.	17 ° 43'8.52 "N	102 ° 48'40.51" E



1.13 Conditions Prior to Project Initiation

The project is a Greenfield waste to energy power project and does not involve generation of GHG emissions for the purpose of their subsequent reduction, removal or destruction. Thus prior to project initiation, there was nothing at site.

In the absence of project activity, the continuation of current practise i.e., generation of equivalent amount of electricity would have been generated from grid connected fossil fuel dominated power plants. Thus, for project activity baseline scenario is electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources. *The municipal solid waste*

that otherwise would have been treated in the project activity would otherwise have been dumped in the landfill sites.

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

The project has received all the necessary approvals for the development and commissioning of the proposed 8MW waste to energy power project from the respective nodal agencies and is in compliance to the local laws and regulations.

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The project is not registered under any other GHG programs. The project is seeking registration only in VCS program.

1.15.2 Projects Rejected by Other GHG Programs

The project is not rejected by any other GHG programs.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

Does the project reduce GHG emissions from activities that are included in an emissions trading program or any other mechanism that includes GHG allowance trading?

☐ Yes ☒ No

If yes, provide the name of the emissions trading program or other mechanism that allows GHG allowance trading.

1.16.2 Other Forms of Environmental Credit

Has the project sought or received another form of GHG-related credit, including renewable energy certificates?

☐ Yes ☒ No

If yes, provide the name of the other program(s) under which the project has sought or received another form of GHG-related credit.

1.17 Sustainable Development Contributions

Social well-being: The project would help in generating employment opportunities during the construction and operation phases. The project activity will lead to development in infrastructure

in the region like development of roads and also may promote business with improved power generation.

Economic well-being: The project is a clean technology investment in the region, which would not have been taken place in the absence of the VCS benefits the project activity will also help to reduce the demand supply gap in the state.

The project activity will generate power using zero emissions waste to energy-based power generation which helps to reduce GHG emissions and specific pollutants like SO_x, NO_x, and SPM associated with the conventional thermal power generation facilities.

Technological well-being: The successful operation of project activity would lead to promotion of Waste to energy based power generation and would encourage other entrepreneurs to participate in similar projects.

Environmental well-being: waste to energy based electricity generation reduces the dependence on fossil fuels and conserves natural resources which are on the verge of depletion. Due to its zero emission the Project activity also helps in avoiding significant amount of GHG emissions.

1.18 Additional Information Relevant to the Project

Leakage Management

Not Applicable

Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

Further Information

Not applicable

2 SAFEGUARDS

2.1 No Net Harm

The project activity involves the construction of waste to energy project. This leads to avoidance of dumping the waste in the landfills. Thus, there is no any significant impact due to implementation of project activity on air, water, soil quality and ambience are envisaged due to the project activity.

2.2 Local Stakeholder Consultation

The details of stakeholder meetings are as follows:

Date of meeting: 06/09/2018

Location: Project site, Thailand

The physical Local stakeholder consultation was carried out by inviting the local stakeholders through public notice. The notice was for all Local stakeholders and not a gender specific. Since project is developed in particular site, for project, the nearby local villagers are most relevant and who are directly and indirectly may be affected. Thus, putting public notice at project site/nearby village involves engagement of all stakeholders for the project activity.

Most of the questions were related to employment opportunities, economic development, benefits from project to villagers and other development activities. The question raised by the villagers are summarised below:

Q: Will there be employment generation due to the project activity for youth from the adjoining areas?

A: Preference would be given for locals in the employment opportunities if there are any.

Q: Will it impact the underground water level in the nearby area?

A: No, it will not impact the underground water level of the nearby area.

2.3 Environmental Impact

The project activity has no significant impact on the environment.

2.4 Public Comments

These details will be updated on the later stages when the project is registered on the standard.

2.5 AFOLU-Specific Safeguards

Not Applicable

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

The United Nations approved consolidated baseline methodologies applicable to this project are Small-scale Methodology: Avoidance of methane production from decay of biomass through controlled combustion, gasification or mechanical/thermal treatment Version 17.01

AMSI.D.: Grid connected renewable electricity generation -- Version 18.02

Following tools have been referred during the estimation of emission reduction calculations as per the above stated methodologies

1. Tool to calculate the emission factor for an electricity system, Version 07.03.
2. Demonstration of additionality of small-scale project activities" Version 13.14
3. Methodological tool "Investment Analysis" Version 11.05
4. Methodological tool "Emissions from solid waste disposal sites" Version 8.06
5. Methodological tool "Project and leakage emissions from transportation of freight7" Version 1.1.0

3.2 Applicability of Methodology

The methodologies AMS IIIE version 17.0 and AMS ID, Version 18.0 is applicable to the project activity under the following conditions:

Applicability Criteria	Applicability status
Applicability conditions of AMS IIIE, Version 17.0	
The project activity does not recover or combust methane unlike AMS-III.G. Nevertheless, the location and characteristics of the disposal site in the baseline condition shall be known, in such a way as to allow the estimation of its methane	The project activity does not combust methane and hence the project activity meets the applicability condition.

1 <https://cdm.unfccc.int/methodologies/DB/GPWCV89KQ7IFPEDCXA92BL6XK7JR3Y>

2 <https://cdm.unfccc.int/methodologies/DB/W3TINZ7KKWCK7L8WTFQQOFQQH4SBK>

3 <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-07-v7.0.pdf>

4 <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-21-v13.1.pdf>

5 <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v11.0.pdf>

6 <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-04-v8.0.pdf>

7 <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-12-v1.1.0.pdf>

emissions.	
If the project activity involves combustion, gasification or mechanical/thermal treatment of partially decayed waste mined (i.e. removed) from a solid waste disposal site in addition to freshly generated waste the project participants shall demonstrate that there is adequate capacity of the combustion, gasification or mechanical/thermal treatment facility to treat the newly generated wastes in addition to the partially decayed wastes removed from the disposal site. Alternately justifications for combusting, gasifying or mechanically/thermally treating the partially decayed wastes instead of the newly generated wastes shall be provided.	The project activity uses both fresh and half decayed wastes and there is sufficient capacity available for both type of the wastes and hence the project activity meets the applicability condition.
If the combustion facility, the produced syngas, producer gas or RDF/SB is used for heat and electricity generation within the project boundary, that component of the project activity may use a corresponding methodology under Type I project activities.	The project activity generates electricity and hence the Methodology AMS ID is being used and hence the project activity meets the applicability condition.
In case of RDF/SB production, project proponents shall provide evidence that no GHG emissions occur, other than biogenic CO₂, due to chemical reactions during the thermal treatment process for example limiting the temperature of thermal treatment to prevent the occurrence of pyrolysis and/or the stack gas analysis.	The project activity does not involve any GHG other than biogenic CO₂ hence the project activity meets the applicability condition.
In case of gasification, the process shall ensure that all the syngas produced, which may contain non-CO₂ GHG, will be combusted and not released unburned to the atmosphere. Measures to avoid physical leakage of the syngas between the gasification and combustion sites shall also be adopted.	The project activity does not produce any form of gas and hence the applicability condition is irrelevant.

In case of RDF/SB processing, the produced RDF/SB should not be stored in such a manner as resulting in high moisture and low aeration favoring anaerobic decay. Project participants shall provide documentation showing that further handling and storage of the produced RDF/SB does not result in anaerobic conditions and do not lead to further absorption of moisture.	The project activity does not store the waste that results in high moisture and hence the applicability condition is satisfied
In case of RDF/SB processing, local regulations do not constrain the establishment of RDF/SB production plants/thermal treatment plants nor the use of RDF/SB as fuel or raw material.	The local conditions does not constrain the establishment of the plant and hence the applicability condition is satisfied.
During the mechanical/thermal treatment to produce RDF/SB no chemical or other additives shall be used.	The project activity does not use any form of additive and hence the applicability condition is satisfied.
In case residual waste from controlled combustion, gasification or mechanical/thermal is stored under anaerobic conditions and/or delivered to a landfill emission from the residual waste shall to be taken into account using the first order decay model (FOD) described in AMS-III.G	The project activity does not produce any residual waste maintained in anaerobic condition and hence the applicability condition is irrelevant.
AMS ID, Version 18.0	
This methodology is applicable to grid-connected renewable power generation project activities that: (a) install Greenfield power plant; (b) involve a capacity addition to (an) existing plant(s); (c) involve a retrofit of (an) existing plant(s)/unit(s); (d) involve a rehabilitation of (an) existing	The proposed project activity is a green field, Indian grid connected renewable power plant. Therefore, it confirms to the said criteria

plant(s)/unit(s); or (e) involve a replacement of (an) existing plant(s)/unit(s)	
The methodology is applicable under the following conditions: The project activity may include renewable energy power plant/unit of one of the following types: hydro power plant/unit with or without reservoir, wind power plant/unit, geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit	The project activity is the installation of a new grid connected renewable Waste to Energy power project. Thus, it meets the first applicability condition
In the case of capacity additions, retrofits, rehabilitations or replacements (except for wind, solar, wave or tidal power capacity addition projects the existing plant/unit started commercial operation prior to the start of a minimum historical reference period of five years, used for the calculation of baseline emissions and defined in the baseline emission section, and no capacity expansion or retrofit or rehabilitation of the plant/unit has been undertaken between the start of this minimum historical reference period and the implementation of the project activity	The proposed project activity is the installation of a new Waste to Energy power plants/units. Therefore, the said criteria is not applicable
In case of hydro power plants, one of the following conditions shall apply: 6. The project activity is implemented in an existing single or multiple reservoirs, with no change in the volume of any of reservoirs; or 7. The project activity is implemented in an existing single or multiple reservoirs, where the volume of the reservoir(s) is increased and the power density calculated using equation (3) is greater than 4 W/m²; or 8. The project activity results in new single or multiple reservoirs and the	The proposed project activity is the installation of Waste to Energy power plants/units. Therefore, the said criteria is not relevant

power density calculate equation (3), is greater than 4 W/m ² .	
If the new unit has both renewable and non-renewable components (e.g. a wind/diesel unit), the eligibility limit of 15 MW for a small-scale CDM project activity applies only to the renewable component. If the new unit co-fires fossil fuel, the capacity of the entire unit shall not exceed the limit of 15 MW	This is a greenfield Waste to Energy electricity generation project and hence the condition is not relevant.
Combined heat and power (co-generation) systems are not eligible under this category.	This is a greenfield Waste to Energy electricity generation project and hence the condition is not relevant.
In the case of project activities that involve the capacity addition of renewable energy generation units at an existing renewable power generation facility, the added capacity of the units added by the project should be lower than 15 MW and should be physically distinct from the existing units.	This is a greenfield Waste to Energy electricity generation project and hence the condition is not relevant.
In the case of retrofit, rehabilitation or replacement, to qualify as a small-scale project, the total output of the retrofitted, rehabilitated or replacement power plant/unit shall not exceed the limit of 15 MW.	The proposed project activity is the installation of Waste to Energy power plants/units. Therefore, the said criteria is not relevant
In the case of landfill gas, waste gas, wastewater treatment and agro-industries projects, recovered methane emissions are eligible under a relevant Type III category. If the recovered methane is used for electricity generation for supply to a grid then the baseline for the electricity component shall be in accordance with procedure prescribed under this methodology. If the recovered methane is used for heat generation or cogeneration other applicable Type-I methodologies such as "AMS-	The proposed project activity is the installation of Waste to Energy power plants/units. Therefore, the said criteria is not relevant

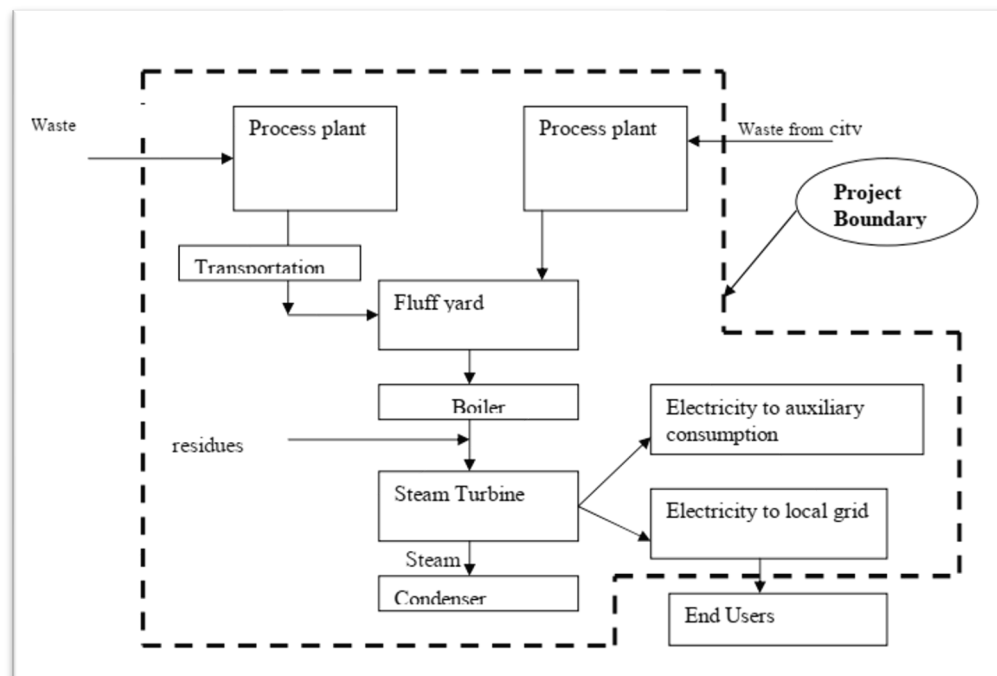
I.C.: Thermal energy production with or without electricity” shall be explored.	
In case biomass is sourced from dedicated plantations, the applicability criteria in the tool “Project emissions from cultivation of biomass” shall apply.	The proposed project activity is the installation of Waste to Energy power plants/units. Therefore, the said criteria is not relevant
In addition, the above applicability conditions, the applicability conditions of tool referred in the methodology AMS ID, Version 18.0 has been referred here under:	
Tool07: Tool to calculate the emission factor for an electricity system Version 7.0	
This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid(e.g. demand-side energy efficiency projects).	The project activity is a greenfield waste to Energy power generation plant and hence, according to the applied methodology, the baseline scenario is electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in “TOOL07: Tool to calculate the emission factor for an electricity system”.
Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, the conditions specified in “Appendix 2: Procedures related to off-grid power generation” should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	Since the project activity is grid connected, this condition is applicable and the emission factor has been calculated accordingly.

In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	The project activity is located in India, a non-Annex I country. Therefore, this criterion is not applicable for the project activity
Under this tool, the value applied to the CO ₂ emission factor of bio fuels is zero	The project activity is a grid connected Waste to Energy power project and therefore, this criterion is not applicable for the project activity
Tool27: Investment analysis version 11.	
This methodological tool is applicable to project activities that apply the methodological tool “Tool for the demonstration and assessment of additionality”, the methodological tool “Combined tool to identify the baseline scenario and demonstrate additionality”, the guidelines “Non-binding best practice examples to demonstrate additionality for SSC project activities”, or baseline and monitoring methodologies that use the investment analysis for the demonstration of additionality and/or the identification of the baseline scenario.	Project activity applies “Non-binding best practice examples to demonstrate additionality for SSC project activities”, Hence this tool is applicable.
In case the applied approved baseline and monitoring methodology contains requirements for the investment analysis that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail.	The project activity is a small-scale grid connected Waste to Energy power project. It refers to the “Non-binding best practice examples to demonstrate additionality for SSC project activities”, Hence this tool is applicable.
Tool21: Demonstration of additionality of small-scale project activities – Version 13.1	
The use of the methodological tool “Demonstration of additionality of small-scale project activities” is not mandatory for project participants when proposing new methodologies. Project participants and coordinating/managing entities may propose alternative methods to demonstrate additionality for consideration by the Executive Board.	Project activity applies the small-scale methodology AMS ID and hence this tool has been considered and applied.

Project participants and coordinating/managing entities may also apply “TOOL19: Demonstration of additionality of microscale project activities” as applicable.	Project activity applies the small-scale methodology AMS ID and hence the Microscale tool is not relevant.
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3.3 Project Boundary

The project boundary includes the waste processing locations, Power plant, sub-stations, grid and all power plants connected to grid. The proposed project activity will evacuate power to the Thailand. Therefore, the entire Thai grid and all connected power plants have been considered in the project boundary for the proposed project activity.



The project does not involve any other emissions sources not foreseen by the methodologies. The greenhouse gases and emission sources included in or excluded from the project boundary are shown in table below.

The table below provides an overview of the emissions sources included or excluded from the project boundary for the determination of baseline and project emissions.

Source		Gas	Included?	Justification/Explanation
Baseline	Source 1 - Waste managed at landfills and Grid	CO ₂	Yes	Main Emission Source
		CH ₄	yes	Main Emission Source

Source		Gas	Included?	Justification/Explanation
	Connected Electricity Generation	N ₂ O	No	Minor Emission source
		Other	No	Not Applicable
Project	Waste is converted into electricity and Greenfield MSW based Power Project activity	CO ₂	No	No CO ₂ emissions are emitted from the project
		CH ₄	No	Project activity does not emit CH ₄
		N ₂ O	No	Project activity does not emit N ₂ O
		Other	No	Not Applicable

3.4 Baseline Scenario

The baseline scenario is the situation where, in the absence of the project activity, biomass and other organic matter is left to decay within the project boundary and methane is emitted to the atmosphere. The baseline emissions are the amount of methane from the decay of the biomass or organic waste treated in the project activity. The yearly methane generation potential is calculated using the first order decay model based on the discrete time estimate method of the IPCC guidelines, as described in category AMS III-E.

The method below is used to evaluate the yearly methane generation potential in the landfill. The quantity of methane projected to be formed during a given year is estimated using a first order decay model based on the discrete time estimate method proposed in the IPCC Guidelines.

MCF of 1 has been used to account for the existence of a suppressed demand situation as described in the “Guidelines on the consideration of suppressed demand in CDM methodologies” as the following conditions apply for the project activity.

- (a) waste is being dumped in an uncontrolled manner in human settlement areas under the current practice due to a lack of organized waste collection and disposal system;
- (b) Only the municipal solid waste is being treated under the project activity and wastes from other sources such as agricultural or agro- industrial wastes are not being treated under the project activity;
- (c) Entire portion of the waste treated under the project activity would comply with the above two conditions.

For the electricity generation part, an Approved small-scale baseline CDM methodology AMS ID, Version 18.0.has been followed along with the “tool to calculate the emission factor for an electricity system, version 7.0.0” are used to establish the baseline scenario.

According to the methodology baseline scenario has been identified as “Electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the “Tool to calculate the emission factor for an electricity system”.

The project activity involved setting up of Waste to Energy Power generation Plant to harness the power of Waste to Energy to produce electricity and supply to the grid. In the absence of the project activity, the equivalent amount of power would have been supplied to the electricity grid by the operation of grid-connected power plants (mainly by fossil fuel fired plants) and by the addition of new generation sources, as reflected in the combined margin (CM) calculations.

Hence, the baseline for the project activity is the equivalent amount of power from the Thailand National grid.

The combined margin ($EF_{grid,CM,y}$) is the result of a weighted average of two emission factor pertaining to the electricity system: the operating margin (OM) and build margin (BM). Calculations for this combined margin must be based on data from an official source (where available) and made publically available

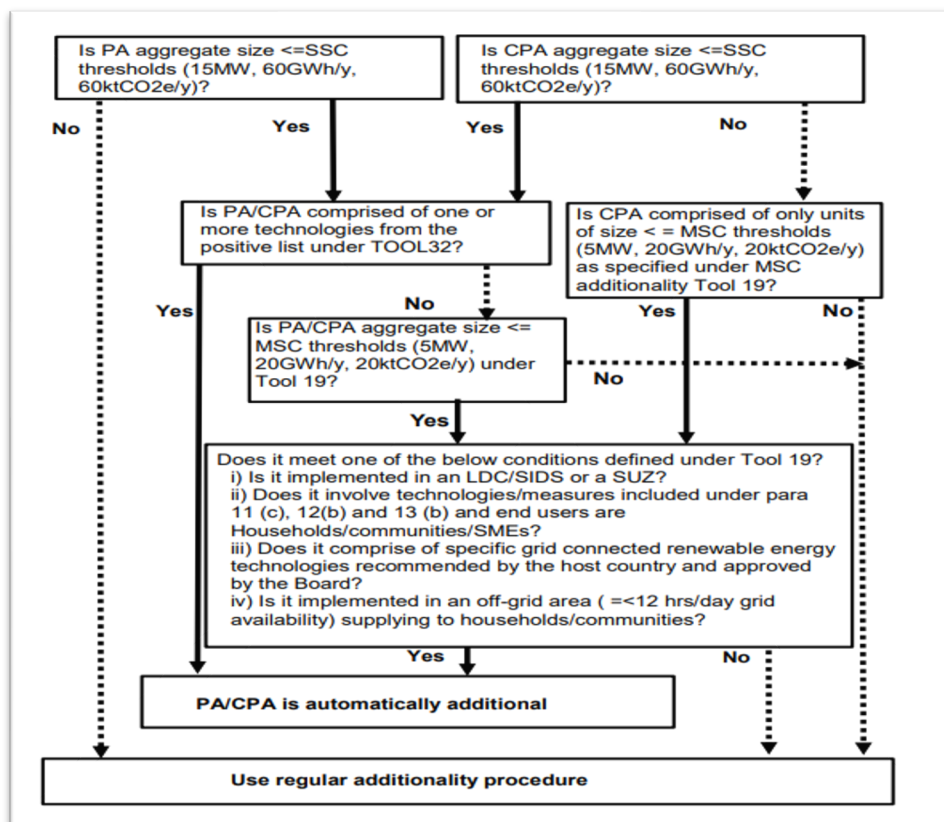
The combined margin of the Thailand national grid used for the project activity is as follows

Parameter	Value	Nomenclature	Source
$EF_{grid,CM,y}$	0.5664 tCO ₂ /MWh	Combined margin CO ₂ emission factor for the project electricity system in year y	Calculated as the weighted average of the operating margin (0.50) & build margin (0.50) values, sourced from Ministry of Energy publication in Thailand.
$EF_{grid,OM,y}$	0.5719 tCO ₂ /MWh	Operating margin CO ₂ emission factor for the project electricity system in year y	Calculated as the 3-year (2014,2015 and 2016) generation-weighted average, sourced from sourced from Ministry of Energy publication in Thailand
$EF_{grid,BM,y}$	0.5609 tCO ₂ /MWh	Build margin CO ₂ emission factor for the project electricity system in year y	Baseline CO ₂ Emission Database, sourced from Ministry of Energy publication in Thailand

The baseline case is in compliance with all applicable legal and regulatory requirements references.

3.5 Additionality

The present VCS project generates power using waste to energy which is a renewable, zero emission source of energy. Baseline considerations for the project are based on approved consolidated baseline methodology AMS IIIE and AMS I.D. The methodology requires the project investor to determine the additionality based on “Methodological Tool- Demonstration of additionality of small-scale project activities”, Version 13.1. The step-wise approach to establish additionality of the project activity has been followed, details of which are provided in the following paragraphs:



Step 1: Is PA aggregate size <= SSC thresholds (15MW, 60GWh/y, 60 ktCO₂e/y)

The size of the proposed project activity is less than 15MW and hence within the SSC thresholds.

Step 2: Is PA/CPA comprised of one or more technologies from the positive list under TOOL 32

This is a waste to energy-based Grid connected Power generation Project. As per the latest version of the Tool 32: Positive List of Technologies” version 04, This technology is not included in the positive list

Step 3: Is PA/CPA Aggregate size <= MSC thresholds (5 MW, 20 GWh/Y, 20 ktCO₂e/y) under Tool19?

Installed capacity of the project is 8 MW *and hence doesn't come under micro activity thresholds.*

Step 4: Does it meet one of the below conditions defined under Tool 19?

1. Is it implemented in an LDC/SIDC or a SUZ?
2. Does it involve technologies/measures included under para 11 (c), 12(b) and 13 (b) and end users are Households/communities/SMEs?
3. Does it comprise of specific grid connected renewable energy technologies recommended by the host country and approved by the Board?
4. Is it implemented in an off-grid area (≤ 12 hrs/day grid availability) supplying to households/communities?

Justification:

The above said conditionalities does not apply and hence the regular additionality approach needs to be followed as stated below.

As per Methodological Tool for the demonstration of Additionality of Small-scale Project Activities (Version 13.1, EB 105, Annex 0418), to establish the project additionality, it has to be shown that the project activity would not have occurred anyway due to at least one of the following barriers:

- Investment barrier: a financially more viable alternative to the project activity would have led to higher emissions;
- Technological barrier: a less technologically advanced alternative to the project activity involves lower risks due to the performance uncertainty or low market share of the new technology adopted for the project activity and so would have led to higher emissions;
- Barrier due to prevailing practice: prevailing practice or existing regulatory or policy requirements would have led to implementation of a technology with higher emissions;
- Other barriers: without the project activity, for another specific reason identified by the project participant, such as institutional barriers or limited information, managerial resources, organizational capacity, financial resources, or capacity to absorb new technologies, emission would have been higher.

The project investor has selected the Investment barrier to demonstrate in a conservative and transparent manner that the proposed GCC project activity is financially unattractive. In line with the guidelines stipulated under Annex 34 of EB 358 (“Non-binding best practice examples to demonstrate additionality for SSC project activities”), a benchmark analysis has been used for the analysis and Equity IRR has been chosen as the financial indicator for the demonstration of the additionality.

Selection of Appropriate Benchmark:

The benchmark has been considered in accordance with CDM Tool 27 “Investment Analysis” Version 11.0 EB 112 Annex 2, “The values in the table in the Appendix may also be used, as a simple default option”.

Methodology deployed for arriving at a suitable value of Benchmark using Default Value has been described below:

1. As the proposed project activity generates power utilizing waste to energy, Group 1 as per para 5a of Appendix of EB 112 Annex 2 has been identified as a suitable category.
2. The investment analysis has been carried out in Nominal terms. Accordingly, Default value as given in table under the Appendix, EB 112 Annex 2 has been adjusted by adding suitable forecasted inflation rate taken from IMF.

In case of inflation data from IMF, Benchmark has been calculated based on inflation rate. As per Para 16 of EB112, Annex 2, the inflation forecast should be for the duration of the crediting period. However, since IMF provides forecast inflation only for 5 year, the project investor has calculated benchmark using 5 year forecast is considered as Benchmark for the project activity.

The benchmark has been computed in the following manner:

Default Value Benchmark:

The cost of equity is determined by selecting the values provided in the table of the Appendix, i.e. Default values for cost of equity (expected return on equity) in the ‘Methodological tool: Investment analysis’.

The Required return on equity (benchmark) was computed in the following manner:

$$\text{Nominal Benchmark}^9 = \{(1 + \text{Real Benchmark}) * (1 + \text{Inflation rate})\} - 1$$

Where:

Default value for Real Benchmark = 12.47% (as per Appendix of EB 112, Annex 2)

8 https://cdm.unfccc.int/Reference/Guidclarif/ssc/methSSC_guid15_v01.pdf

1. 9 As per Pg. 320 of Corporate Finance, Second Edition of Aswath Damodaran

Inflation Rate forecast for by IMF for Thailand¹⁰.

Benchmark estimation:

The Cost of Equity has been considered using the “Methodological tool: Investment analysis” available at the time of decision making as well as the latest available value. As a conservative approach, the minimum value of benchmark has been considered as calculated using these 2 approaches.

Table under Appendix in EB 112, Annex 2 specifies default value of expected return on equity in real terms for Energy Industries (Group 1) in Thailand = 12.47%¹¹

Thus, minimum cost of equity considered for calculation of Benchmark = 12.47%

Inflation Forecast for India as per IMF website¹² and corresponding benchmark values applicable at the time of investment decision time:

Project	Inflation Forecast	Benchmark
	5 Years	5 Years
Project activity 1	1.73%	12.47%

Hence a benchmark of 12.47% has been selected for both the project activities.

Calculation and comparison of financial indicators

The period considered for Post Tax Equity IRR calculations is 25 years, which corresponds to the operational lifetime of the project activity.

Depreciation, and other non-cash items related to the project activity, which have been deducted in estimating gross profits on which tax is calculated, is added back to net profits for the purpose of calculating the financial indicator.

Input values considered for the IRR calculation are provided below.

Particulars	Value	Unit	Source/Remarks
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¹⁰ <https://www.imf.org/en/Publications/WEO/Issues/2016/12/31/Uneven-Growth-Short-and-Long-Term-Factors>

¹¹ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v10.0.pdf>

¹² <https://www.imf.org/en/Publications/WEO/weo-database/2019/April/weo-report?c=534,&s=PCIPCH,&sy=2018&ey=2024&ssm=0&scsm=1&sc=0&ssd=1&ssc=0&sic=0&sort=country&ds=.&br=1>

Capacity of the project	8	MW	From DPR, Appendix 2
Plant Load Factor	70.00%	%	From DPR, Appendix 2
Annual Net generation	44.15	GWh	Calculated
Project cost	1455.00	THB Million	From DPR, Appendix 2
Debt	75%	%	From DPR, Appendix 2
Equity	25%	%	As per DPR
Debt	1091.25	THB Million	Calculated
Equity	363.75	THB Million	Calculated
Interest rate	4.00%	%	From DPR, Appendix 2
Debt Repayment tenure	8	years	From DPR, Appendix 2
Moratorium	1	year	From DPR, Appendix 2
Fuel cost	200	Baht/Tonnes	From DPR, Appendix 2
Fuel cost escalation	2%		From DPR, Appendix 2
Fuel consumption per annum	90,450	Tonnes/year	From DPR, Appendix 2
Operation and Maintenance	15.5	THB Million	From DPR, Appendix 2
Escalation in O & M	3%	%	From DPR, Appendix 2
Service tax on O & M fees	0.00%	%	As per prevailing tax rates
Insurance & overhead	3.64	THB Million	As per quotation received from Insurance companies for this project activity
Escalation in Insurance	5%		From DPR, Appendix 2
Other expenses (Lease + Standby + Admin)	0.50%		As per DPR
Tariff – First 8 years	5.78	THB/kWh	Tariff Negotiations
Tariff - After 8 years	5.08	THB/kWh	Tariff Negotiations
Depreciation Rate (Book)	4.00%	%	From DPR, Appendix 2
Income tax rate (1-8 years)	0.00%	%	

Income tax rate (9th year onwards)	20.00%	%	Calculated
Salvage Value	10%	%	

Post Tax Equity IRR for the project activities against the benchmark values are shown in table below. Thus, it is evident that the project is not financially attractive as the equity IRR is less below the benchmark value.

Post tax Equity IRR	3.07%
Benchmark Value	12.47%

Sensitivity Analysis:

The robustness of the conclusion drawn above, namely that the project is not financially attractive, has been tested by subjecting critical assumptions to reasonable variation. As required by Annex 02 of EB 112, only variables, including the initial investment cost, that constitute more than 20% of either total project costs or total project revenues should be subjected to reasonable variation. PP has identified the total revenue from the project activity is dependent on the Tariff, Plant Load Factor, Project Cost, Fuel cost and O&M Costs constitute more than 20% of the project costs. These factors have been subjected to a 10% variation on either side and the results of the sensitivity analysis indicate that even after applying such variation the EIRR does not cross the benchmark.

Variation %	-10%	Normal	10%	Variation required to reach benchmark	Value required to reach benchmark
Tariff	1.49%	3.07%	4.70%	52.00%	8.79 TBH/kWh
PLF	1.49%	3.07%	4.70%	52.00%	106.40%
Fuel Cost	3.22%	3.07%	2.91%	NA	NA
Project Cost	4.54%	3.07%	1.91%	-39.00%	887.55 TBH(Mn)
O&M Cost	3.22%	3.07%	2.92%	NA	NA

An analysis has been done to identify the percentage variation at which the financial indicators will equal/breach the benchmark and the probability of its occurrence. Based on sensitivity analysis it can be concluded that the proposed project activity is additional with reasonable variation in values and is not likely to reach the benchmark value. The occurrence of these events is unlikely for the following reasons:

1. **Tariff:** The Tariff rate of electricity used for investment analysis i.e., 5.78 TBH/kWh is sourced from the PPA negotiations communications applicable at the time of investment decision. Furthermore, the project will breach the benchmark value at a tariff variation of 52%. However, the actual tariff based on the PPAs signed is 5.78 TBH/kWh fixed for first 7 years and 5.08 TBH/KWh for the remaining period of the project lifetime of 25 years which is much below the tariff value required to breach the benchmark value. Hence, 52% increase in tariff is unlikely.
2. **PLF:** The PLF value considered is based on TEV which is taken from the Third Party PLF report i.e. 70% and the IRR breach the benchmark value at a PLF variation of more than 52%. The increase in PLF value to breach the benchmark is highly unlikely as the generation at the site has been substantially less and equity IRR at the observed PLF values are less than the benchmark value and given the analysis above it's highly unlikely that PLF will increase above breaching value.
3. **Project Cost:** The project cost considered for investment analysis i.e. 1455 million TBH. The cost is sourced from TEV which is based on the negotiations with Supplier. A variation of -39.00% is required for IRR to breach benchmark which is not possible as the project is almost ready to be commissioned. The actual cost incurred in commissioning of the project is same as the estimated cost. Hence reduction of 39.00% is unlikely to happen.
4. **Fuel Cost:** The project cost considered for investment analysis i.e. 200 Baht/Tonnes. The cost is sourced from TEV which is based on the negotiations with Supplier. A variation is not possible as the fuel supply contract is already in place. The actual cost incurred for the fuel of the project is same as the estimated cost. Hence reduction is unlikely to happen.
5. **O&M Costs:** The sensitivity analysis reveals that O&M will breach the benchmark at negative values and is hypothetical case. Since the O&M cost is subject to escalation (as evidence by the O&M agreement) and also subject to inflationary pressure, any reduction in the O&M costs is highly unlikely. The O&M contract has been executed at 15.5 Mn THB/ year at which also the equity IRR is much below the benchmark value.

Conclusion:

As described above, the project fulfils all necessary requirements of additionality specified in the 'Methodological Tool for the demonstration of Additionality of Small-scale Project Activities (Version 13.1, EB 105,). Hence, the project is additional.

3.6 Methodology Deviations

There is no methodology deviations

4 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

Emission Reductions:

$$ER_y = ER_y(\text{MSW}) + ER_y(\text{Elec})$$

Where

ER_y = Emission reductions in year y (t CO₂e/yr)

$ER_y(\text{MSW})$ = Emission reduction from methane avoidance in year y (t CO₂e/yr)

$ER_y(\text{Elec})$ = Emission reduction from Electricity generation in year y (t CO₂e/yr)

As per the paragraph 43 of the methodology AMS ID Version 18.0 emission reductions are calculated as follows

Emission Reductions from Electricity:

$$ER_y(\text{Elec}) = BE_y - PE_y - LE_y$$

Where

$ER_y(\text{Elec})$ = Emission reductions in year y (t CO₂e/yr)

BE_y = Baseline emissions in year y (t CO₂/yr)

PE_y = Project emissions in year y (t CO₂/yr)

LE_y = Leakage emissions in year y (t CO₂/yr)

Baseline Emissions:

As per the approved consolidated Methodology AMS ID version 18.0 that Baseline emissions include only CO₂ emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants. The baseline emissions are to be calculated as follows:

$$BE_y = EG_{PJ,y} \times EF_{\text{grid,CM},y}$$

Where,

BE_y = Baseline emissions in year y (t CO₂/yr)

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)

$EF_{grid,CM,y}$ = Combined margin CO₂ emission factor for grid connected power generation in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system” (t CO₂/MWh)

AS per para 26 of AMS ID, version 18.0, when the project activity is installation of Greenfield power plant, then:

$$EG_{PJ,y} = EG_{facility,y}$$

Where,

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)

$EG_{facility,y}$ = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr)

The $EG_{facility,y}$ is estimated from the PLF provided as per the third party engineering company report as below:

Project activity	$EG_{facility,y}$ (MWh/yr)
8 MW Nongkhai Waste to Energy Project in Thailand	39,917

As per the methodology combined margin grid emission factor has been calculated as per the “Tool to calculate the grid emission factor for an Electricity System” version 07.

Thailand Grid Emission Factor for GHG Reduction Project/Activity dated 28/09/2017 Published by Thailand Greenhouse Gas Management Organization (Public Organization)¹³ has been used for the calculation of emission factor.

As per the "Tool to calculate the emission factor for an electricity system" Version 07.0, EB 100, Annex 4, the following steps have been followed.

- (a) **Step 1:** Identify the relevant electricity systems;
- (b) **Step 2:** Choose whether to include off-grid power plants in the project electricity system (optional);
- (c) **Step 3:** Select a method to determine the operating margin (OM);
- (d) **Step 4:** Calculate the operating margin emission factor according to the selected method;

¹³ http://ghgreduction.tgo.or.th/images/Grid_Emission_Factor_2559_-_Finalised.pdf

- (e) **Step 5:** Calculate the build margin (BM) emission factor;
- (f) **Step 6:** Calculate the combined margin (CM) emission factor.

Step 1: Identify the relevant electricity systems

For the purpose of determining the electricity emission factor, the project electricity system is defined as the electricity transmission system of Thailand which is a single system connected by transmission lines throughout the country and owned by the Electricity Generating Authority of Thailand (EGAT). Electricity imports from a connected electricity system are included and as per Tool to calculate emission factor of an electricity system, for the purpose of determining the operating margin emission factor, 0 tCO₂/MWh is applied.

Step 2: Choose whether to include off-grid power plants in the project electricity system (optional)

Only grid connected power plants are included in the calculation, as per Option I of the “Tool to calculate the emission factor for an electricity system” version 07.0.0

Step 3: Select a method to determine the operating margin (OM)

The calculation of the operating margin emission factor (EF_{grid,OM,y}) is based on one of the following

methods, which are described under Step 4:

- (a) Simple OM; or
- (b) Simple adjusted OM; or
- (c) Dispatch data analysis OM; or
- (d) Average OM.

The simple OM method (Option a) is used for this study as the low-cost/must-run resources (LC/MR) constitute less than 50% of total electricity production. The share of low-cost/must run power plants in the last two years are in the range of 4.55% to 6.69 as shown in the below table.

ปี พ.ศ.	การผลิต พลังงานไฟฟ้ารวม (GWh)	LC/MR Analysis			
		Hydro	RE	Total LC/MR	% of LC/MR
2555	166,446	8,431	2,701	11,132	6.69
2556	164,826	5,412	3,427	8,839	5.36
2557	168,685	5,164	3,993	9,157	5.43
2558	169,040	3,724	4,230	7,954	4.71
2559	169,168	3,019	4,685	7,704	4.55

Data Source: EGAT(2017)

As per tool to calculate emission factor for an electricity system (Version 07), The simple OM method (option a) can only be used if low-cost/must-run resources constitute less than 50% of total grid generation in: 1) average of the five most recent years, or 2) based on long-term

averages for hydroelectricity production. Since the low cost/must run resources constitute less than 50% of total grid generation as seen from the average of five most recent years, the Simple OM method can be used to calculate the Operating Margin Emission factor.

PP has chosen ex ante option, thus, no monitoring and recalculation of the emissions factor during the crediting period is required. PP has considered a data vintage of 3-year generation-weighted average, based on the most recent data available at the time of submission of the CDM-PDD to the verifier for validation.

Step 4: Calculate the operating margin emission factor ($EF_{grid,OMSimple,y}$) according to the selected method

The simple OM emission factor is calculated as the generation-weighted average CO₂ emissions per unit net electricity generation (tCO₂/MWh) of all generating power plants serving the system, not including low-cost / must-run power plants / units.

The simple OM may be calculated:

Option A: Based on the net electricity generation and a CO₂ emission factor of each power unit;

Or

Option B: Based on the total net electricity generation of all power plants serving the system and the fuel types and total fuel consumption of the project electricity system.

Thailand Greenhouse Gas Management Organization (Public Organization) has published the Thailand Grid Emission Factor for GHG Reduction Project/Activity on 28/09/2017 based on detailed authenticated information obtained from EGAT. This provides information about the Combined Margin Emission Factor of Thailand national grid. The Combined Margin is calculated ex ante using the guidelines provided by the UNFCCC in the “Tool to calculate the emission factor for an electricity system, Version 07”. We have, therefore, used the Combined Margin data published in the Grid Emission Factor for GHG Reduction Project/Activity, for calculating the Baseline Emission Factor.

As per “Tool to calculate the emission factor for an electricity system”, Option B (“Calculation based on total fuel consumption and electricity generation of the system”) is used to calculate simple OM emission factor. Where Option B is used, the simple OM emission factor is calculated based on the net electricity supplied to the grid by all power plants serving the system, not including lowcost/must-run power plants/units, and based on the fuel type(s) and total fuel consumption of the project electricity system, as follows:

$$EF_{grid,OMsimple,y} = \frac{\sum_i FC_{i,y} \times NCV_{i,y} \times EF_{CO2,i,y}}{EG_y}$$

Where:

$EF_{grid,OMsimple,y}$ Simple operating margin CO2 emission factor in year y (tCO2/MWh)

$EG_{m,y}$ - Net quantity of electricity generated and delivered to the grid by power unit m in year y (MWh)

$FC_{i,y}$ - Amount of fuel type i consumed in the project electricity system in year y (mass or volume unit)

$NCV_{i,y}$ - Net calorific value (energy content) of fuel type i in year y (GJ/mass or volume unit)

$EF_{CO2,i,y}$ - CO2 emission factor of fuel type i in year y (t CO2/GJ)

EG_y - Net electricity generated and delivered to the grid by all power sources serving the system, not including low-cost/must-run power plants/units, in year y (MWh)

i - All fuel types combusted in power sources in the project electricity system in year y

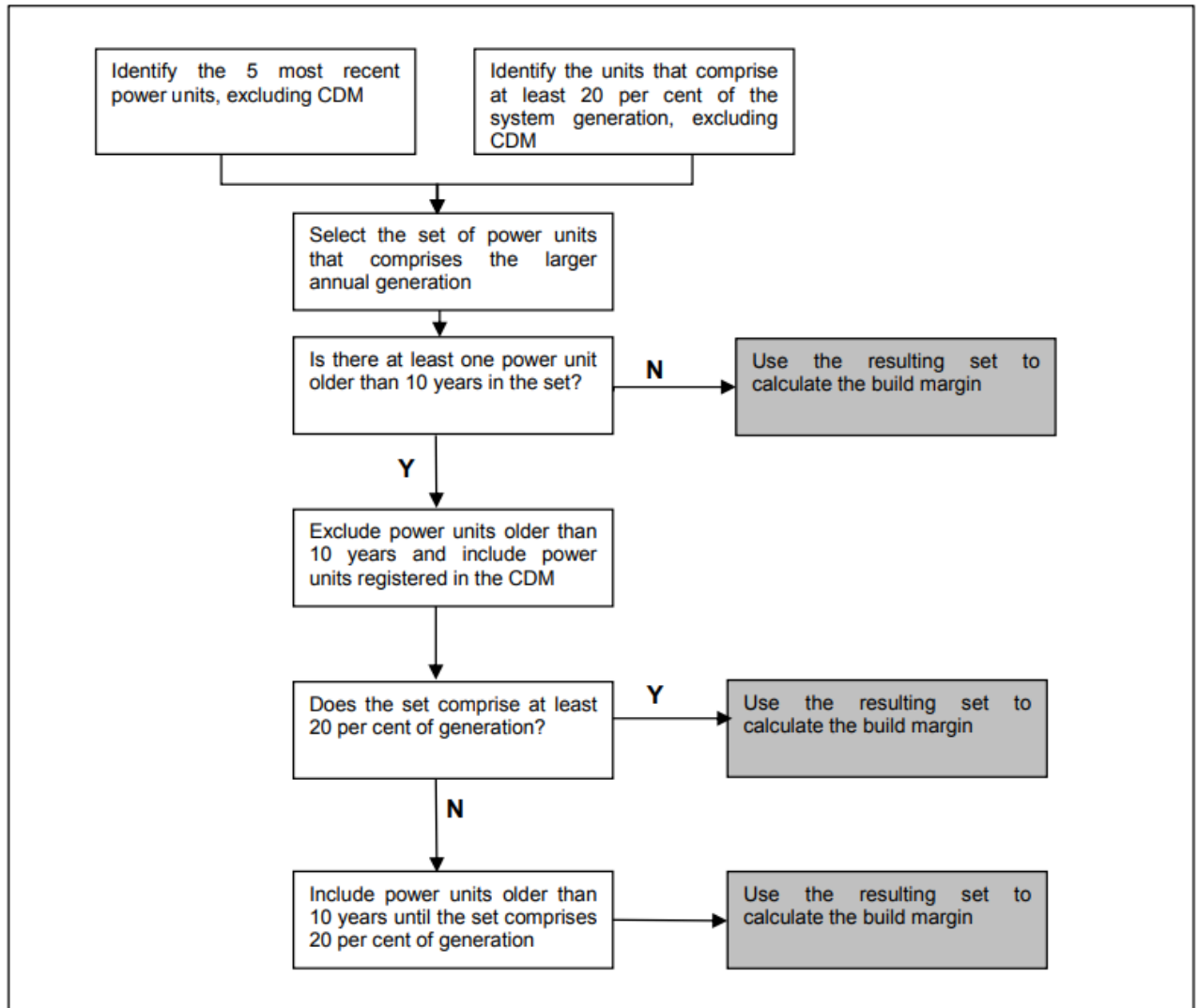
y- the relevant year as per the data vintage chosen in STEP 3

As per the Thailand emission factor database published on 28th September 2017 weighted average operating margin is as below:

$EF_{om,y} = 0.5719 \text{ tCO}_2/\text{MWh}$

STEP 5: Calculate the build margin emission factor ($EF_{BM,y}$)

The project participants have chosen Option I, i.e., fixing build margin emission factor ex ante based on the most recent information available on units already built for sample group m at the time of CDM PDD submission to the verifier for validation. The build margin emissions factor is the generation-weighted average emission factor (tCO2/MWh) of a sample group of power units, during the most recent year y for which power generation data is available. The Sample group of power units m used to calculate the build margin should be determined via the procedure summarized in the diagram of the Tool



Following this procedure, AEGSET >/20% is larger than AEGSET 5-units and all of these power units started supplying electricity to the grid less than 10 years ago, therefore AEGSET >/20% is applied as power units m for the Build Margin Using the equation given in the step 5 for the OM calculation., the Built margin is calculated for the year 2016 is as below:

$$EF_{BM,y} = 0.5609 \text{ tCO}_2/\text{MWh}$$

STEP 6: Calculate the combined margin (CM) emissions factor

The combined margin is the weighted average of the simple operating Margin and the build margin. In particular, for intermittent and non-dispatch able generation types such as wind and solar photovoltaic, the Tool to calculate the emission factor for an electricity system, Version 07.0.0, EB 100, Annex 4, allows to weigh the operating margin and Build margin at 75% and 25%, respectively for wind and solar projects and 50% and 50%, respectively for hydro and biomass projects.

The baseline emission factor is calculated using the combined margin approach as described in the following steps:

Calculation of Baseline Emission Factor EF_y

The baseline emission factor EF_y is calculated as the weighted average of the Operating Margin emission factor ($EFOM,y$) and the Build Margin emission factor ($EFBM,y$):

$$EF_y = wOM * EFOM,y + wBM * EFBM,y$$

Where,

WOM 50% weight for WTE projects

WBM 50% weight for WTE projects

EFOM,y calculated as described in Steps 3&4 above (tCO₂/MWh)

EFBM,y calculated as described in Steps 5 above (tCO₂/MWh)

$$\begin{aligned} \text{Baseline Emission factor (Thailand national grid)} &= 0.5 * 0.5719 + 0.5 * 0.5609 \\ &= 0.5664 \text{ tCO}_2/\text{MWh} \end{aligned}$$

The baseline emission factor is ex-ante parameter and will remain constant throughout the crediting period.

$EF_{grid,,}$	=	Combined Margin Grid Emission Factor = 0.5664 tCO ₂ /MWh
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Baseline emissions for the Methane part

$$BE_y = BE_{CH_4,SWDS,y}$$

Where:

BE_y = Baseline emissions at year y during crediting period (t CO₂e)

$BE_{CH_4,SWDS,y}$ = Yearly Methane Generation Potential of the wastes diverted to be disposed in the landfill from the beginning of the project (x=1) up to the year y, calculated according to the “Tool to determine methane emissions avoided from disposal of waste at a solid waste disposal site” (t CO₂e)

The simplified formulae as per the appendix to the Methodological Tool 04, Version 8.0 provides for

$$BE_{CH_4,SWDS,y} = \varphi_y \times (1 - f_y) \times GWP_{CH_4} \times \sum_{x=1}^y Default_x \times W_x$$

$BE_{CH_4,SWDS,y}$ = Baseline, project or leakage methane emissions occurring in year y generated from waste disposal at a SWDS during a time period ending in year y (t CO₂e/yr)

x = Years in the time period in which waste is disposed at the SWDS, extending from the first year in the time period (x = 1) to year y (x = y)

y = Year of the crediting period for which methane emissions are calculated (y is a consecutive period of 12 months)

W_x = Amount of solid waste disposed or prevented from disposal in the SWDS in the year x (t)

φ_y = Model correction factor to account for model uncertainties for year y

f_y = Fraction of methane captured at the SWDS and flared, combusted or used in another manner that prevents the emissions of methane to the atmosphere in year y

GWP_{CH_4} = Global Warming Potential of methane

4.2 Project Emissions

As per the approved consolidated Methodology AMS ID: “For most renewable energy power generation project activities, $PE_y = 0$. However, some project activities may involve project emissions that can be significant. These emissions shall be accounted for as project emissions by using the following equation:

$$PE_y = PE_{FF,y} + PE_{GP,y} + PE_{HP,y}$$

PE_y = Project emissions in year y (t CO₂e/yr)

$PE_{FF,y}$ = Project emissions from fossil fuel consumption in year y (t CO₂/yr)

$PE_{GP,y}$ = Project emissions from the operation of dry, flash steam or binary geothermal power plants in year y (t CO₂e/yr)

$PE_{HP,y}$ = Project emissions from water reservoirs of hydro power plants in year y (tCO₂e/yr)

As the project activity is the installation of a new grid-connected Waste to Energy Power plant and does not involve any project emissions from fossil fuel, operation of dry, flash steam or binary geothermal power plants, and from water reservoirs of hydro power plants. Therefore $PE_{FF,y}$, $PE_{GP,y}$, $PE_{HP,y}$ are equal to zero and thus, $PE_y = 0$.

For the MSW part in the methodology

$$PE_y = PE_{y,comb} + PE_{y,transp} + PE_{y,power}$$

PE_y = Project activity direct emissions in the year y (t CO₂e)

$PE_{y,comb}$ = Emissions through combustion and gasification of non-biomass carbon of waste and RDF/SB in the year y (t CO₂e)

$PE_{y,transp}$ = Emissions through incremental transportation in the year y (t CO₂e)

$PE_{y,power}$ = Emissions through electricity or diesel consumption in the year y (t CO₂e)

$$PE_{y,comb} = Q_{y,non-biomass} \times 44/12 + Q_{y,fuel} \times EF_{y,fuel}$$

Where:

$Q_{y,non\ biomass}$ = Non-biomass carbon of the waste and RDF/SB combusted/gasified in the year y (tonnes of carbon)

$Q_{y,fuel}$ = Quantity of auxiliary fossil fuel used in the year y (tonnes)

$EF_{y,fuel}$ = CO₂ emission factor for the combustion of the auxiliary fossil fuel (tonnes CO₂ per tonne fuel, according to latest IPCC Guidelines)

Project emissions from trucks for incremental collection activities will be estimated following the methodological tool "Project and leakage emissions from transportation of freight".

4.3 Leakage

No other leakage emissions are considered. The emissions potentially arising due to activities such as power plant construction and upstream emissions from fossil fuel use (e.g. extraction, processing, transport etc.) are neglected.

Hence Emission reductions will be calculated as per the below equation

$$ER_y = BE_y = EG_{PJ,y} \times EF_{grid,CM,y}$$

4.4 Net GHG Emission Reductions and Removals

The ex-ante emission reductions (ERy) for the project activity are calculated as follows

$$ERy = BEy - PEy - Ley$$

Where,

ERy = Emission Reduction in tCO₂/year

BEy = Baseline emission in tCO₂/year

PEy = Project emissions in tCO₂/year

LEy = Leakage Emissions in tCO₂/year

Year	Estimated baseline emissions or removals (tCO ₂ e)	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated net GHG emission reductions or removals (tCO ₂ e)
year 1	31973	1202	0	30771
Year 2	38774	1202	0	37571
year 3	43767	1202	0	42565
year 4	47441	1202	0	46238
year 5	50116	1202	0	48913
year 6	52050	1202	0	50848
year 7	53450	1202	0	52247
Total	3,117,571	8,417	0	309,154

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	EF _{grid,OM,y}
Data unit	tCO ₂ /MWh
Description	Operating Margin CO ₂ emission factor in year y of Thailand Grid
Source of data	Report „Thailand Grid Emission Factor for GHG Reduction Project/Activity“ dated 28/09/2017 Published by Thailand Greenhouse Gas Management Organisation (Public Organisation)
Value applied	0.5719
Justification of choice of data or description of measurement methods and procedures applied	Calculated in line with “Tool to calculate the emission factor for an electricity system” using data from Ministry of Energy publication The value used is calculated ex-ante as generation based weighted average of last three years of the operating margin provided in the database. Weighted average $= \frac{\sum_{i=1 \text{ to } n} (\text{Net generation in operating margin in year } i * \text{Simple operating margin in year } i)}{\sum_{i=1 \text{ to } n} (\text{Net generation in operating margin of year } i)}$
Purpose of Data	Baseline Emission calculation
Comments	The operating margin emission factor is a 3-year generation-weighted average (2014-2016). The operating Margin is calculated ex ante and fixed during the crediting period

Data / Parameter	EF _{grid,BM,y}
Data unit	tCO ₂ /MWh
Description	Report „Thailand Grid Emission Factor for GHG Reduction Project/Activity“ dated 28/09/2017 Published by Thailand Greenhouse Gas Management Organisation (Public Organisation)
Source of data	Ministry of Energy publication

Value applied	0.5609
Justification of choice of data or description of measurement methods and procedures applied	Calculated in line with “<i>Tool to calculate the emission factor for an electricity system</i>” using data from Ministry of Energy publication The value used is calculated ex-ante as generation based weighted average of last three years of data provided in the database.
Purpose of Data	Baseline Emission calculation
Comments	The Build Margin would be calculated ex ante and fixed during the crediting period. For ex ante calculation the most recent data 2016 available has been used and the build margin is thus calculated.

Data / Parameter	$EF_{grid,CM,y}$
Data unit	tCO ₂ /MWh
Description	Operating Margin CO ₂ emission factor in year y of Thailand Grid.
Source of data	Report „Thailand Grid Emission Factor for GHG Reduction Project/Activity” dated 28/09/2017 Published by Thailand Greenhouse Gas Management Organisation (Public Organisation)
Value applied	0.5664
Justification of choice of data or description of measurement methods and procedures applied	Calculated in line with “<i>Tool to calculate the emission factor for an electricity system</i>” using data from Ministry of Energy publication The value used is calculated ex-ante as generation based weighted average of last three years of data provided in the database.
Purpose of Data	Baseline Emission calculation
Comments	The combined margin would be calculated ex-ante and fixed for the entire crediting period.

Data / Parameter	Φ default									
Data unit										
Description	Default value for the model correction factor to account for model uncertainties									
Source of data	Default									
Value applied	For project or leakage emissions: $\phi_{\text{default}} = 1$. For baseline emissions: refer to the table below to identify the appropriate factor based on the application of the tool (A or B) and the climate where the SWDS is located. Default values for the model correction factor <table><tr><th></th><th>Humid/wet conditions</th><th>Dry conditions</th></tr><tr><td>Application A</td><td>0.75</td><td>0.75</td></tr><tr><td>Application B</td><td>0.85</td><td>0.80</td></tr></table>		Humid/wet conditions	Dry conditions	Application A	0.75	0.75	Application B	0.85	0.80
	Humid/wet conditions	Dry conditions								
Application A	0.75	0.75								
Application B	0.85	0.80								
Justification of choice of data or description of measurement methods and procedures applied	-									
Purpose of Data	-									
Comments	The table above is applicable to Option 1 in the procedure “Determining the model correction factor (ϕ)”									

Data / Parameter	F
Data unit	-
Description	Fraction of methane in the SWDS gas (volume fraction)
Source of data	IPCC 2006 Guidelines for National Greenhouse Gas Inventories
Value applied	0.5
Justification of choice of data or description of	-

measurement methods and procedures applied	
Purpose of Data	-
Comments	Upon biodegradation, organic material is converted to a mixture of methane and carbon dioxide

5.2 Data and Parameters Monitored

Data / Parameter	EG _{facility,y}
Data unit	MWh/Year
Description	Quantity of net electricity generation supplied by the project (Waste to Energy) plant/unit to the grid in year y
Source of data	Measured & Calculated
Description of measurement methods and procedures to be applied	Credit note monthly generation report from the electricity board/DISCOM
Frequency of monitoring/recording	Continuous monitoring & monthly recording
Value applied	39917

Monitoring equipment	Energy meters of accuracy class 0.2s.																		
	Main and Check meters are installed at the govt substations by the electricity utility to measure the net exported electricity from																		
	<table><tr><td></td><td>Main Mater</td><td>Check Meter</td></tr><tr><td>Type of meter</td><td colspan="2">Bi-directional meters</td></tr><tr><td>Location of meter</td><td colspan="2">Substation</td></tr><tr><td>Accuracy of meter</td><td colspan="2">0.2s</td></tr><tr><td>Serial number of meter</td><td>216439211</td><td>216449211</td></tr><tr><td>Calibration frequency</td><td colspan="2">Once in 5 years</td></tr></table>		Main Mater	Check Meter	Type of meter	Bi-directional meters		Location of meter	Substation		Accuracy of meter	0.2s		Serial number of meter	216439211	216449211	Calibration frequency	Once in 5 years	
		Main Mater	Check Meter																
	Type of meter	Bi-directional meters																	
	Location of meter	Substation																	
	Accuracy of meter	0.2s																	
	Serial number of meter	216439211	216449211																
Calibration frequency	Once in 5 years																		
the plant.																			
QA/QC procedures to be applied	The meter(s) shall be calibrated and maintained by the National utility as per their own schedule, and this frequency of meter calibration is not within the control of the Project Proponent. Calibration of electricity meters is carried out in-line with the Nation standard which recommends at least once in 5 year calibration or whenever abnormal difference/inconsistency is observed between main meter and check meter.																		
Purpose of data	Baseline Emission Calculations																		
Calculation method	The value of net electricity generation supplied to the grid as invoice forms the basis for calculation of the emission reductions; which can be with the onsite monitoring records The Net electricity is calculated based on Export- import Monthly meter readings are taken from the main and check meter installed at metering point and certified by the representatives of Electricity Officials and the representatives of the project participant. The Joint Meter Reports is cross checked with the Invoices.																		
Comments	-																		

Data / Parameter	Wx
Data unit	t
Description	Total amount of waste disposed in a SWDS in year x or month i
Source of data	Measurements by project participants
Description of measurement methods and procedures to be applied	Measured & Calculated
Frequency of monitoring/recording	Measurement: Continuous Recording: Monthly
Value applied	90450
Monitoring equipment	Measure on wet basis
QA/QC procedures to be applied	The weigh bridge will be calibrated once every two years as per the monitoring plan laid out by the project participants.
Purpose of data	Baseline Emission Calculations.
Calculation method	-
Comments	-

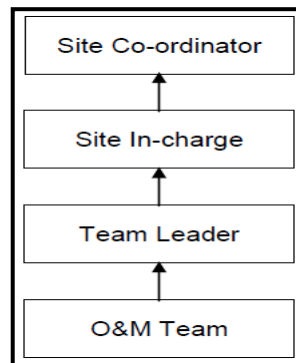
5.3 Monitoring Plan

The monitoring plan is developed in accordance with the methodology AMS III E, version 17.0 and AMS ID , Ver 18.0 . The monitoring plan, which will be implemented by the project owner, describes about the monitoring organization, parameters to be monitored, monitoring practices, quality assurance, quality control procedures, data storage and archiving.

The project proponent has entered into agreement with the O&M Contractor for the operation and maintenance of the Waste to Energy projects. O&M contractor will provide a monthly report, which includes generation data, major breakdown events and machine availability. Project manager is responsible for recording of monthly meter readings of export and import. Monthly power export and import data will be sent regularly to site in charge of each project separately

Monitoring roles and responsibilities:

The data for the project is compiled by the O&M Contractor and subsequently stored by the Project Owners, the reporting and data flows as per the below mentioned flow chart starting from Site O&M team which monitors day to day operational data and monthly recording. The reporting responsibilities for the project are described below;



The Site In-charge will be responsible for carrying out internal auditing and QA/QC. All the values from generation record will be checked with the invoices for consistency. In case there are any non-conformances identified. The Site In-charge will investigate the error and revise the record to correct it. In any case where values have slightest of variation in different records the most conservative value will be taken in the project monitoring report.

Personal Training:

The project employs qualified and experienced persons for plant operation. The training period shall be for three months, as this would be adequate and necessary to ensure proper imparting of the objective. The training course will be thoroughly and meticulously designed, highlighting the objectives, salient features, operational aspects and trouble shooting.

Emergency preparedness:

In case Main meter or Check meter is found to be outside the acceptable limits of accuracy or faulty or not functioning properly, it will be repaired, recalibrated or replaced as soon as possible. In the event that the Main meter is not in service as a result of maintenance, repairs or testing, the Check meter will be used for readings

Data recording & archiving: The project proponent shall maintain data both in electronic form and hard copies. The monitored data shall be archived till 2 years after the completion of crediting period.